

Reverse-check review package — VLQ_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	VLQ/model/VLQ_gen.fr
original model name	VLQ_gen (hidden from the agent)
paper	VLQ/text/hep-ph_0607115.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LyTP (=)

```
-lamT (bar[Q3L].Phi.tR) - lamTp (bar[Q3L].Phi.tpR) + HC[-lamT (bar[Q3L].Phi.tR) - lamTp  
↪ (bar[Q3L].Phi.tpR)]
```

LDTP (=)

```
-MT0 bar[tpL].tpR - MTmix bar[tpL].tR + HC[-MT0 bar[tpL].tpR - MTmix bar[tpL].tR]
```

LWTP (=)

```
-(gw/Sqrt[2]) (cT CKM[3, ff] bar[t].Ga[mu].ProjM.dq[ff] + sT CKM[3, ff]  
↪ bar[tp].Ga[mu].ProjM.dq[ff]) W[mu] + HC[-(gw/Sqrt[2]) (cT CKM[3, ff]  
↪ bar[t].Ga[mu].ProjM.dq[ff] + sT CKM[3, ff] bar[tp].Ga[mu].ProjM.dq[ff]) W[mu]]
```

LZTP (=)

```
-(gw/(2 cw)) (cT^2 bar[t].Ga[mu].ProjM.t + sT cT bar[t].Ga[mu].ProjM.tp + sT cT  
↪ bar[tp].Ga[mu].ProjM.t + sT^2 bar[tp].Ga[mu].ProjM.tp) Z[mu]
```

LHTP (=)

```
(gw/(2 MW)) ((cT^2 MT bar[t].ProjP.t + sT cT MTP bar[t].ProjP.tp + sT cT MT bar[tp].ProjP.t  
↪ + sT^2 MTP bar[tp].ProjP.tp) H + HC[(cT^2 MT bar[t].ProjP.t + sT cT MTP bar[t].ProjP.tp  
↪ + sT cT MT bar[tp].ProjP.t + sT^2 MTP bar[tp].ProjP.tp) H])
```

LyBP (=)

```
-lamB (bar[Q3L].Phibar.bR) - lamBp (bar[Q3L].Phibar.bpR) + HC[-lamB (bar[Q3L].Phibar.bR) -  
↪ lamBp (bar[Q3L].Phibar.bpR)]
```

LDBP (=)

```
-MB0 bar[bpL].bpR - MBmix bar[bpL].bR + HC[-MB0 bar[bpL].bpR - MBmix bar[bpL].bR]
```

LWBP (=)

```
-(gw/Sqrt[2]) (cB CKM[ff,3] bar[uq[ff]].Ga[mu].ProjM.b + sB CKM[ff,3]
  ↳ bar[uq[ff]].Ga[mu].ProjM.bp) W[mu] + HC[-(gw/Sqrt[2]) (cB CKM[ff,3]
  ↳ bar[uq[ff]].Ga[mu].ProjM.b + sB CKM[ff,3] bar[uq[ff]].Ga[mu].ProjM.bp) W[mu]]
```

LZBP (=)

```
-(gw/(2 cw)) (cB^2 bar[b].Ga[mu].ProjM.b + sB cB bar[b].Ga[mu].ProjM.bp + sB cB
  ↳ bar[bp].Ga[mu].ProjM.b + sB^2 bar[bp].Ga[mu].ProjM.bp) Z[mu]
```

L4CKM (=)

```
-(gw/Sqrt[2]) ((cU4 CKM[3, ff] - sU4 sV4 CKM[2, ff] - sU4 cV4 sW4 CKM[1, ff])
  ↳ bar[t].Ga[mu].ProjM.dq[ff] + (sU4 CKM[3, ff] + cU4 sV4 CKM[2, ff] + cU4 cV4 sW4 CKM[1,
  ↳ ff]) bar[t4].Ga[mu].ProjM.dq[ff] + (cV4 CKM[2, ff] - sV4 sW4 CKM[1, ff])
  ↳ bar[c].Ga[mu].ProjM.dq[ff] + cW4 CKM[1, ff] bar[u].Ga[mu].ProjM.dq[ff]) W[mu] +
  ↳ HC[-(gw/Sqrt[2]) ((cU4 CKM[3, ff] - sU4 sV4 CKM[2, ff] - sU4 cV4 sW4 CKM[1, ff])
  ↳ bar[t].Ga[mu].ProjM.dq[ff] + (sU4 CKM[3, ff] + cU4 sV4 CKM[2, ff] + cU4 cV4 sW4 CKM[1,
  ↳ ff]) bar[t4].Ga[mu].ProjM.dq[ff] + (cV4 CKM[2, ff] - sV4 sW4 CKM[1, ff])
  ↳ bar[c].Ga[mu].ProjM.dq[ff] + cW4 CKM[1, ff] bar[u].Ga[mu].ProjM.dq[ff]) W[mu]]
```

L4Mass (=)

```
-MT4 bar[t4].t4 - MB4 bar[b4].b4 - MN4 bar[n4].n4 - ME4 bar[e4].e4
```

Blank-slate reconstruction

Reconstruction from `sanitized.fr`

Conventions

The file uses

$$P_L \equiv \text{ProjM} = \frac{1 - \gamma^5}{2}, \quad P_R \equiv \text{ProjP} = \frac{1 + \gamma^5}{2}.$$

Hermitian conjugates are written as `HC[...]`. Charge conjugation `CC`, covariant derivatives `DC`, and field strengths `FS` do not appear in this file.

Repeated flavor index f is summed over the three SM generations. I write

$$V_{ij} \equiv \text{CKM}[i, j], \quad d_f \equiv \text{dq}[ff].$$

The file uses `W[mu]` in charged-current terms of the form $\bar{u} \gamma^\mu P_L d W_\mu$; by electric charge conservation this is W_μ^+ , with the hermitian conjugate giving the W_μ^- interaction.

For the electroweak-doublet Yukawa contractions,

$$Q_{3L} = \begin{pmatrix} t_L \\ b_L \end{pmatrix}.$$

Gauge invariance of the written terms implies that `Phi` is the up-type Higgs doublet contraction, i.e. $\tilde{H} = i\sigma^2 H^*$, while `Phibar` is the down-type Higgs doublet H . The dot `.` denotes contraction of the $SU(2)_L$ doublet index and color indices where present.

No explicit `DC[...]` kinetic terms are present. If canonical kinetic terms are assumed for a declared fermion ψ , the gauge content follows from the declared color and charges:

$$D_\mu \psi = \left[\partial_\mu + ig_s G_\mu^A T_3^A + ie Q_\psi A_\mu + i \frac{g_w}{c_w} (T_\psi^3 - Q_\psi s_w^2) Z_\mu + \text{charged } W^\pm \text{ terms if an } SU(2)_L \text{ multiplet is specified} \right] \psi.$$

In the actual class declarations here, no explicit $SU(2)_L$ index is assigned to any new field.

Lagrangian

LyTP

$$\mathcal{L}_{\text{LyTP}} = -\lambda_T \bar{Q}_{3L i} \tilde{H}^i t_R - \lambda'_T \bar{Q}_{3L i} \tilde{H}^i t'_R + \text{h.c.}$$

Here $t' \equiv \text{tp}$. Color indices are contracted between $\bar{Q}_{3L}$ and t_R, t'_R .

LDTP

$$\mathcal{L}_{\text{LDTP}} = -M_{T0} \bar{t}'_L t'_R - M_{T\text{mix}} \bar{t}'_L t_R + \text{h.c.}$$

LWTP

$$\mathcal{L}_{\text{LWTP}} = -\frac{g_w}{\sqrt{2}} [c_T V_{3f} \bar{t} \gamma^\mu P_L d_f + s_T V_{3f} \bar{t}' \gamma^\mu P_L d_f] W_\mu^+ + \text{h.c.}$$

with

$$s_T = \sin \theta_T, \quad c_T = \cos \theta_T.$$

LZTP

$$\mathcal{L}_{\text{LZTP}} = -\frac{g_w}{2c_w} [c_T^2 \bar{t} \gamma^\mu P_L t + s_T c_T \bar{t} \gamma^\mu P_L t' + s_T c_T \bar{t}' \gamma^\mu P_L t + s_T^2 \bar{t}' \gamma^\mu P_L t'] Z_\mu$$

LHTP

$$\mathcal{L}_{\text{LHTP}} = \frac{g_w}{2M_W} [c_T^2 M_T \bar{t} P_R t + s_T c_T M_{T'} \bar{t} P_R t' + s_T c_T M_T \bar{t}' P_R t + s_T^2 M_{T'} \bar{t}' P_R t'] h + \text{h.c.}$$

Here $h \equiv \text{H}$, $M_T \equiv \text{MT}$, and $M_{T'} \equiv \text{MTP}$.

LyBP

$$\mathcal{L}_{\text{LyBP}} = -\lambda_B \bar{Q}_{3L i} H^i b_R - \lambda'_B \bar{Q}_{3L i} H^i b'_R + \text{h.c.}$$

Here $b' \equiv \text{bp}$.

LDBP

$$\mathcal{L}_{\text{LDBP}} = -M_{B0} \bar{b}'_L b'_R - M_{B\text{mix}} \bar{b}'_L b_R + \text{h.c.}$$

LWBP

$$\mathcal{L}_{\text{LWBP}} = -\frac{g_w}{\sqrt{2}} [c_B V_{f3} \bar{u}_f \gamma^\mu P_L b + s_B V_{f3} \bar{u}_f \gamma^\mu P_L b'] W_\mu^+ + \text{h.c.}$$

with

$$s_B = \sin \theta_B, \quad c_B = \cos \theta_B.$$

LZBP

$$\mathcal{L}_{\text{LZBP}} = -\frac{g_w}{2c_w} [c_B^2 \bar{b} \gamma^\mu P_L b + s_B c_B \bar{b} \gamma^\mu P_L b' + s_B c_B \bar{b}' \gamma^\mu P_L b + s_B^2 \bar{b}' \gamma^\mu P_L b'] Z_\mu$$

L4CKM

$$\mathcal{L}_{\text{L4CKM}} = -\frac{g_w}{\sqrt{2}} \left[(c_{U4}V_{3f} - s_{U4}s_{V4}V_{2f} - s_{U4}c_{V4}s_{W4}V_{1f}) \bar{t}\gamma^\mu P_L d_f \right. \\ \left. + (s_{U4}V_{3f} + c_{U4}s_{V4}V_{2f} + c_{U4}c_{V4}s_{W4}V_{1f}) \bar{t}_4\gamma^\mu P_L d_f \right. \\ \left. + (c_{V4}V_{2f} - s_{V4}s_{W4}V_{1f}) \bar{c}\gamma^\mu P_L d_f \right. \\ \left. + c_{W4}V_{1f} \bar{u}\gamma^\mu P_L d_f \right] W_\mu^+ + \text{h.c.}$$

where

$$s_{U4} = \sin \theta_{U4}, \quad c_{U4} = \cos \theta_{U4}, \quad s_{V4} = \sin \theta_{V4}, \quad c_{V4} = \cos \theta_{V4}, \quad s_{W4} = \sin \theta_{W4}, \quad c_{W4} = \cos \theta_{W4}.$$

L4Mass

$$\mathcal{L}_{\text{L4Mass}} = -M_{T4} \bar{t}_4 t_4 - M_{B4} \bar{b}_4 b_4 - M_{N4} \bar{n}_4 n_4 - M_{E4} \bar{e}_4 e_4$$

Field table

Class	Symbol	Spin	$SU(3)_c$ rep	$SU(2)_L$ rep	Q	Y	Self- conjugate	Mass
F[100]	tp	1/2 Dirac fermion	3	not declared	+2/3	+2/3	no	MTP = 600.
F[101]	bp	1/2 Dirac fermion	3	not declared	-1/3	-1/3	no	MBP = 600.
F[102]	t4	1/2 Dirac fermion	3	not declared	+2/3	+1/6	no	MT4 = 600.
F[103]	b4	1/2 Dirac fermion	3	not declared	-1/3	+1/6	no	MB4 = 600.
F[104]	n4	1/2 Dirac fermion	1	not declared	0	-1/2	no	MN4 = 100.
F[105]	e4	1/2 Dirac fermion	1	not declared	-1	-1/2	no	ME4 = 100.

External parameters

Parameter	Value	Appears in	Meaning
thT	0.1	LWTP, LZTP, LHTP through s_T, c_T	Left-chiral mixing angle between the SM top and tp in charged, neutral, and Higgs interactions.
lamT	1.	LyTP	Yukawa coupling multiplying $\bar{Q}_{3L}\tilde{H}t_R$.
lamTp	1.	LyTP	Yukawa coupling multiplying $\bar{Q}_{3L}\tilde{H}t'_R$.
MT0	600.	LDTP	Vectorlike Dirac mass parameter for tp .
MTmix	0.	LDTP	Mass-mixing parameter multiplying $\bar{t}'_L t_R$.

Parameter	Value	Appears in	Meaning
thB	0.01	LWBP, LZBP through s_B, c_B	Left-chiral mixing angle between the SM bottom and bp in charged and neutral interactions.
lamB	1.	LyBP	Yukawa coupling multiplying $\bar{Q}_{3L} H b_R$.
lamBp	1.	LyBP	Yukawa coupling multiplying $\bar{Q}_{3L} H b'_R$.
MB0	600.	LDBP	Vectorlike Dirac mass parameter for bp .
MBmix	0.	LDBP	Mass-mixing parameter multiplying $\bar{b}'_L b_R$.
thU4	0.1	L4CKM through s_{U4}, c_{U4}	Mixing angle controlling the redistribution of charged-current couplings between t and t_4 .
thV4	0.1	L4CKM through s_{V4}, c_{V4}	Mixing angle modifying charged-current couplings involving t, t_4, c .
thW4	0.	L4CKM through s_{W4}, c_{W4}	Mixing angle modifying charged-current couplings involving u, c, t, t_4 .

Physics summary

The file encodes an extension of the SM fermion sector with heavy Dirac quarks **tp** and **bp** that mix chirally with the top and bottom sectors, plus additional Dirac fermions **t4**, **b4**, **n4**, and **e4** with explicit Dirac masses. The interactions modify $W^\pm$, Z , and Higgs couplings of the third-generation quarks and introduce charged-current mixing between t_4 and SM down-type quarks.

It mediates processes such as heavy-quark production and decay through $t' \rightarrow W d_f, Z t, h t$, $b' \rightarrow W u_f, Z b$, and charged-current production or decay involving t_4 through the extended CKM-like structure.

Paper cross-check

Paper Locations

The paper defines the vector-like top model in Sec. 2.1, “The case for a vector-like t' quark”: charged current Eq. (7), mixing rotation Eq. (8), rescaled CKM entries Eqs. (9)-(10), Yukawa terms Eq. (11), Dirac mass terms Eq. (12), neutral-current and Higgs FCNC structures Eqs. (14)-(16). It briefly comments on the analogous down-type vector-like b' model immediately after Eq. (16). The fourth-generation model is defined in Sec. 2.2, “The case for a fourth generation”: a unitary $V_{4 \times 4}$ matrix with rotations Eq. (31), and entries Eqs. (32)-(35). The paper also notes anomaly cancellation requiring fourth-generation leptons near Sec. 2.2.1, after Eq. (42), but does not write their Lagrangian.

paper term (eq. ref)	reconstruction term	verdict	notes
Vector-like t' field content: one $Q = +2/3$ vector-like electroweak iso-singlet quark, Sec. 2.1	tp : Dirac color triplet, $Q = +2/3$, no explicit $SU(2)_L$ index	agree	The absence of an $SU(2)_L$ index is consistent with the paper's iso-singlet t' .
Charged current $L_{W^\pm} =$ $-g/\sqrt{2}[\bar{u}_L V \gamma^\mu d_L W_\mu^+ +$ $h.c.]$, Eq. (7), with $V_{ti} = V_{ti}^{(0)} \cos \theta$, $V_{t'i} = V_{ti}^{(0)} \sin \theta$, Eqs. (9)-(10)	LWTP : $-g_w/\sqrt{2}[c_T V_{3f} \bar{t} \gamma^\mu P_L d_f +$ $s_T V_{3f} \bar{t}' \gamma^\mu P_L d_f] W_\mu^+ +$ $h.c.$	agree	Matches the t, t' rows of Eq. (7) after the rotation. The reconstruction omits unchanged u, c charged currents, which is acceptable if it is only listing non-SM or modified terms.
Left-handed $t - t'$ rotation $R_{34}(\theta)$, Eq. (8)	$s_T = \sin \theta_T$, $c_T =$ $\cos \theta_T$ in LWTP , LZTP , LHTP	agree	Same physical mixing structure.
Yukawa terms $L_y(t') =$ $\lambda(t^0, b^0)_L \Phi t_R^0 +$ $\lambda'(t^0, b^0)_L \Phi t_R'^0 + h.c.$, Eq. (11)	LyTP : $-\lambda_T \bar{Q}_{3L} \tilde{H} t_R -$ $\lambda'_T \bar{Q}_{3L} \tilde{H} t'_R + h.c.$	agree	Same gauge-invariant content for an up-type Yukawa, modulo sign and notation. The reconstruction's $\tilde{H}$ is the expected up-type contraction.
Dirac mass terms $L_D(t') =$ $M \bar{t}_L^0 t_R^0 + M' \bar{t}_L'^0 t_R^0 + h.c.$, Eq. (12)	LDTP : $-M_{T0} \bar{t}_L' t'_R -$ $M_{Tmix} \bar{t}_L' t_R + h.c.$	agree	Same vector-like mass and mixing mass content; overall sign is a convention.
Neutral current $L_{Z^0} =$ $-g/(2c_w) \bar{u}_L V V^\dagger \gamma^\mu u_L Z_\mu^0$, Eq. (14), with $V V^\dagger$ block in Eq. (16)	LZTP : $-g_w/(2c_w)[c_T^2 \bar{t} P_L t +$ $s_T c_T \bar{t} P_L t' +$ $s_T c_T \bar{t}' P_L t + s_T^2 \bar{t}' P_L t'] Z_\mu$	agree	Matches the t, t' block of $V V^\dagger$ in Eq. (16).
Higgs FCNC $L_{H^0} =$ $g/(2M_W)[\bar{u}_L V V^\dagger M_u u_R +$ $h.c.]\bar{H}^0$, Eq. (15), with $M_u =$ $\text{diag}(m_u, m_c, m_t, m_{t'})$, Eq. (16)	LHTP : $g_w/(2M_W)[c_T^2 M_T \bar{t} P_R t +$ $s_T c_T M_{T'} \bar{t} P_R t' +$ $s_T c_T M_T \bar{t}' P_R t +$ $s_T^2 M_{T'} \bar{t}' P_R t'] h + h.c.$	agree	Correct mass appears by right-handed flavor column, as in $V V^\dagger M_u$.
Down-type vector-like b' model: paper says only that the 3×4 matrix is the transpose analogue of Eq. (6), with $V_{tb} = V_{tb}^{(0)} \cos \theta_d$, immediately after Eq. (16)	LyBP , LDBP , LWBP , LZBP	extra-in-reconstruction	The paper mentions the analogous b' model but does not define its full Yukawa, mass, Z , or Higgs Lagrangian. The reconstruction includes a full explicit down-type vector-like model.
Down-type vector-like Yukawa analogue, not explicitly written in paper	LyBP : $-\lambda_B \bar{Q}_{3L} H b_R -$ $\lambda'_B \bar{Q}_{3L} H b'_R + h.c.$	extra-in-reconstruction	Plausible analogue of Eq. (11), but not a paper equation.

paper term (eq. ref)	reconstruction term	verdict	notes
Down-type vector-like Dirac mass analogue, not explicitly written in paper	LDBP: $-M_{B0}\bar{b}'_L b'_R - M_{Bmix}\bar{b}'_L b_R + h.c.$	extra-in-reconstruction	Plausible analogue of Eq. (12), but not defined in the paper.
Down-type vector-like charged current transpose analogue after Eq. (16)	LWBP: $-g_w/\sqrt{2}[c_B V_{f3}\bar{u}_f\gamma^\mu P_L b + s_B V_{f3}\bar{u}_f\gamma^\mu P_L b']W_\mu^+ + h.c.$	agree	This agrees with the stated transpose analogue if one treats the paper's comment as defining the b' case. It is still extra relative to the paper's main models.
Down-type vector-like neutral current analogue, not explicitly written in paper	LZBP: $-g_w/(2c_w)[c_B^2\bar{b}P_L b + s_B c_B\bar{b}P_L b' + s_B c_B\bar{b}'P_L b + s_B^2\bar{b}'P_L b']Z_\mu$	extra-in-reconstruction	Expected by analogy, but the paper does not write this term.
Fourth-generation model: unitary $V_{4\times 4}$, no tree-level hadronic Z^0 FCNCs, Sec. 2.2	t4, b4, n4, e4 Dirac fields plus L4CKM	agree	The reconstruction captures the presence of extra up/down quarks and leptons, but its field-representation table is incomplete/problematic.
Fourth-generation CKM parametrization $V_{4\times 4} = R_{34}(\theta_u)R_{24}(\theta_v)R_{14}(\theta_w) \begin{pmatrix} V_{3\times 3}^{(0)} & 0 \\ 0 & 1 \end{pmatrix}$, Eq. (31)	L4CKM with s_{U4}, s_{V4}, s_{W4} and c_{U4}, c_{V4}, c_{W4}	agree	Same three-angle parametrization.
$V_{ui} = c_w V_{ui}^{(0)}$, Eq. (32)	L4CKM : $c_{W4}V_{1f}\bar{u}\gamma^\mu P_L d_f$	agree	Matches Eq. (32), identifying $W4 \leftrightarrow w$.
$V_{ci} = c_v V_{ci}^{(0)} - s_v s_w V_{ui}^{(0)}$, Eq. (33)	L4CKM : $(c_{V4}V_{2f} - s_{V4}s_{W4}V_{1f})\bar{c}\gamma^\mu P_L d_f$	agree	Matches Eq. (33).
$V_{ti} = c_u V_{ti}^{(0)} - s_u s_v V_{ci}^{(0)} - s_u c_v s_w V_{ui}^{(0)}$, Eq. (34)	L4CKM : $(c_{U4}V_{3f} - s_{U4}s_{V4}V_{2f} - s_{U4}c_{V4}s_{W4}V_{1f})\bar{t}\gamma^\mu P_L d_f$	agree	Matches Eq. (34).
$V_{t'i} = s_u V_{ti}^{(0)} + c_u s_v V_{ci}^{(0)} + c_u c_v s_w V_{ui}^{(0)}$, Eq. (35)	L4CKM : $(s_{U4}V_{3f} + c_{U4}s_{V4}V_{2f} + c_{U4}c_{V4}s_{W4}V_{1f})\bar{t}'\gamma^\mu P_L d_f$	agree	Matches Eq. (35), with reconstruction's t4 corresponding to the fourth-generation t' .

paper term (eq. ref)	reconstruction term	verdict	notes
Fourth-generation quark electroweak structure: a sequential fourth generation with t'_L, b'_L in an $SU(2)_L$ doublet is implied by Sec. 2.2 and the unitary CKM construction	Reconstruction field table lists t4 and b4 as Dirac fields with no declared $SU(2)_L$ rep; both have $Y = +1/6$	disagree	A sequential fourth generation requires chiral electroweak assignments: left-handed doublet $Y = 1/6$, right-handed singlets $Y = 2/3, -1/3$. A Dirac field with no $SU(2)_L$ rep and a single listed $Y = 1/6$ is not a complete or gauge-covariant representation statement.
Fourth-generation lepton content required by anomaly cancellation, discussed after Eq. (42)	n4, e4 Dirac fields with no declared $SU(2)_L$ rep; both have $Y = -1/2$	disagree	The paper only states the need for leptons; a sequential generation would have a left-handed lepton doublet $Y = -1/2$ plus right-handed singlets. The reconstruction's single Dirac-field hypercharge assignment is incomplete.
Fourth-generation masses: paper uses $m_{t'}$, discusses $m_{b'} = m_{t'}$ near Eq. (40), but does not write gauge-invariant mass/Yukawa terms	L4Mass: $-M_{T4}\bar{t}_4t_4 - M_{B4}\bar{b}_4b_4 -$ $M_{N4}\bar{n}_4n_4 - M_{E4}\bar{e}_4e_4$	disagree	These look like explicit vector-like Dirac masses. For a chiral sequential fourth generation, bare Dirac masses are not $SU(2)_L \times U(1)_Y$ invariant before EWSB; masses should arise from Higgs Yukawa terms, even if the phenomenology later treats physical masses as inputs.
Fourth-generation neutral-current structure: paper says tree-level hadronic Z^0 FCNCs are forbidden in Sec. 2.2	No LZ4 FCNC term in reconstruction	agree	Absence of fourth-generation Z -FCNC terms is consistent with the paper.

paper term (eq. ref)	reconstruction term	verdict	notes
Fourth-generation Yukawa sector for t', b' and leptons	No fourth-generation Higgs/Yukawa terms in reconstruction	missing-in-reconstruction	The paper does not explicitly write these terms, but a sequential fourth generation under the SM gauge group requires them for gauge-invariant masses. This is missing if the reconstruction is meant to be a Lagrangian-level model, rather than a phenomenological mass/CKM implementation.

Disagreements To Check

issue	severity	what a human should check
The reconstruction includes a full vector-like b' model (LyBP, LDBP, LWBP, LZBP) while the paper only briefly comments on an analogous down-type case after Eq. (16).	substantive	Check whether the implementation intentionally bundled the paper's aside as an additional optional model, or whether the target paper model should only include the vector-like up-type t' and fourth-generation cases.
The fourth-generation fields are listed as Dirac fields with no $SU(2)_L$ representation, while the paper's Sec. 2.2 implies a sequential chiral fourth generation.	substantive	Check the original implementation's class declarations and gauge quantum numbers, especially whether left- and right-handed components are represented separately elsewhere.
t4 and b4 both have listed $Y = +1/6$, and n4/e4 both have listed $Y = -1/2$, without separating left doublets from right singlets.	substantive	Verify whether the reconstruction confused doublet hypercharge with full Dirac-field hypercharge, since right-handed fourth-generation singlets need different hypercharges.
L4Mass uses explicit Dirac mass terms for fourth-generation fermions.	substantive	Check whether these are merely post-EWSB physical mass terms in a phenomenological implementation, or intended as gauge-invariant pre-EWSB Lagrangian terms; the latter would not match a chiral sequential fourth generation.

issue	severity	what a human should check
The reconstruction omits explicit fourth-generation Yukawa interactions.	substantive	Check whether the implementation only works in the mass basis for phenomenology, or whether a gauge-invariant Lagrangian reconstruction was expected.
Overall signs in the reconstructed Yukawa and Dirac mass terms differ from the signs printed in Eqs. (11)-(12).	convention	Check the implementation's Lagrangian sign convention; this likely has no physical effect if used consistently.
The reconstruction's vector-like top charged current omits unchanged u, c SM charged currents from Eq. (7).	cosmetic	Check whether the reconstruction is intentionally listing only modified/new interactions rather than the full SM charged-current Lagrangian.

Overall Assessment

The reconstruction matches the paper's vector-like up-type t' model well at the level of charged-current mixing, Yukawa structure, vector-like mass mixing, and the Z/H FCNC coefficient structure in Eqs. (7)-(16). Its fourth-generation charged-current block also correctly reproduces the three-angle CKM parametrization of Eqs. (31)-(35). The main caveats are that the reconstruction contains an explicit vector-like b' sector that the paper only mentions as an aside, and its fourth-generation field and mass descriptions look more like phenomenological Dirac fields than a gauge-invariant sequential fourth-generation Lagrangian. A human reviewer should therefore distinguish between “implementation-level mass-basis phenomenology” and “full electroweak-gauge-invariant model definition” before interpreting the mismatches.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 29 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: [] approve [] approve with corrections [] reject
- Notes: